**Q1** A $130\ \Omega$ resistor and a $30\ \mu F$ capacitor are connected in parallel across a $200\ V$, $50\ Hz$ supply. Calculate:

(a) The current in each branch
(b) The resultant current
(c) The phase difference between the resultant current and the applied voltage

Sketch the phasor diagram.

**Q2** A resistor and a capacitor are connected in series across a $150\ V$ a.c. supply. When the frequency is $40\ Hz$ the current is $5\ A$, and when the frequency is $50\ Hz$ the current is $6\ A$.

a) Find the resistance and capacitance of the resistor and capacitor respectively
b) If they are now connected in parallel across the $150\ V$ supply, find the total current and its phase angle when the frequency is $50\ Hz$.

**Q3** A $31.8\ mF$ capacitor, a $127.5\ mH$ inductor of resistance $30\ \Omega$ and a $100\ \Omega$ resistor are all connected in parallel to a $200\ V$, $50\ Hz$ supply. Calculate the current in each branch. Find the total current and its phase angle. Draw a phasor diagram to show these currents.

**Q4** A parallel network consists of two branches A and B. Branch A has a resistance of $10\ \Omega$ and an inductance of $0.1\ H$ in series. Branch B has a resistance of $20\ \Omega$ and a capacitance of $100\ \mu F$ in series. The network is connected to a single-phase supply of $230\ V$ at $50\ Hz$. Calculate the magnitude and phase angle of the current taken from the supply. Draw the phasor diagram.

*End*